

*MA.8.NSO.1.4***Benchmark**

MA.8.NSO.1.4 Express numbers in scientific notation to represent and approximate very large or very small quantities. Determine how many times larger or smaller one number is compared to a second number.

Example: Roderick is comparing two numbers shown in scientific notation on his calculator. The first number was displayed as $2.3147E27$ and the second number was displayed as $3.5982E - 5$. Roderick determines that the first number is about 10^{32} times bigger than the second number.

Connecting Benchmarks/Horizontal Alignment

- MA.7.NSO.1.1
- MA.8.NSO.1.5, MA.8.NSO.1.6
- MA.8.GR.1.2

Terms from the K-12 Glossary

- Exponent
- Scientific Notation

Vertical Alignment**Previous Benchmarks**

- MA.7.NSO.2.1

Next Benchmarks

- MA.912.GR.4.5

Purpose and Instructional Strategies

In elementary school, students began to explore the place value system by understanding a number's value is ten times larger than the number to its right and $\frac{1}{10}$ of the number to its left using whole numbers. In grade 7 accelerated, students develop an understanding of Laws of Exponents (Appendix E) with numerical expressions. Students also focus on generating equivalent numerical expressions with whole-number exponents and rational number bases. Additionally, students use the knowledge of Laws of Exponents to work with scientific notation. In Geometry, students will solve problems involving density in terms of area and volume which can be represented using scientific notation when the numbers are large. Additionally, students can apply their scientific notation knowledge in science courses.

- Instruction builds students' number sense with scientific notation. Students should see how representing numbers in a given form allows for students to see the magnitude of the number in an efficient way.

- Instruction connects place value and expanded form with scientific notation. This will allow students to compare very large and very small numbers concisely.

thousands	hundreds	tens	ones	.	tenths	hundredths	thousandths
10^3	10^2	10^1	10^0	.	10^{-1}	10^{-2}	10^{-3}
3	2	4	0	.	0	0	0
			0	.	3	2	4

Scientific notation for the numbers within the chart would be represented as 3.24×10^3 and 3.24×10^{-1} respectively.

- Students should use place value knowledge to determine how many times larger a number is compared to another. Students should develop patterns to conclude that if the exponent increases by one, the value increases 10 times, as well as if the exponent decreases by one, the value decreases 10 times.
 - For example, if students are determining how many times bigger 7×10^9 is than 3×10^8 . Students will need to recognize that 7 is approximately 2 times larger than 3, and 10^9 is 10 times greater than 10^8 . Therefore, to determine how many times greater, a student would reason that 7×10^9 is approximately 2×10 (or 20) times greater than 3×10^8 .
- Instruction connects student's understanding of scientific notation to choosing appropriate units of measures.
- When using calculators to represent very large and very small numbers with an exponent indicated as “E”, instruction relates the number following “E” as the power of 10.

Common Misconceptions or Errors

- Students often confuse the meaning of the exponent and the value of the number in scientific notation.
- Some students misrepresent scientific notation by not expressing the number as a product of a power of 10 and a number that is at least 1 and less than 10.
- Students may incorrectly interpret the “E” on a calculator display as an error message.
- Students may interpret the comparison of two numbers in scientific notation incorrectly.
 - For example, if students were asked what is 3 times larger than 3×10^3 , they may respond with 9×10^9 instead of the correct response of 9×10^3 .
 - For example, if a student determines the first number is 10^4 times bigger than the second number, they may incorrectly believe the first number is 4 times as big as the second number instead of 10,000 times bigger.

Strategies to Support Tiered Instruction

- Instruction includes making connections of a number written in standard form to the same number written in scientific notation. Key connections include recognizing the similarities in the first two digits of both numbers and the connections between the place value of the number in standard form and the exponent of the power.
- Teacher provides opportunities for students to utilize calculators and provides instruction on the various calculator notations for scientific notation.
- Instruction includes rewriting whole numbers in scientific notation when finding products or quotients with scientific notation to demonstrate the correct use of operations and laws of exponents.
 - For example, if the student is asked what is five times larger than 2×10^4 , they should be multiplying 5×2 , and not multiply by the exponent.
- Teacher provides opportunities for students to check their work by rewriting numbers in standard form and applying any necessary operations before comparing their solution to the solution found with the use of a calculator.
- Instruction includes the use of manipulatives such as base 10 blocks to make connections to the purpose of utilizing scientific notation.
 - For example, the teacher could pose the question: “What would be the best way for us to represent 2430 using Base Ten Blocks. We could use 2430 individual Base Ten Unit blocks, or we could use 2- Base Ten Cubes, 4 Base Ten Flats, and 3 Base Ten Rods. Students can then see that it would be easier to represent 2430 using the Cubes, Flats, and Rods as opposed to the large amount of individual Unit blocks. When students see how it would be easier to use the larger blocks to represent the number, the teacher can explain how it is similar to using scientific notation to write out very large or very small numbers. Instead of writing 2873000000000000000, they can write 2.873×10^{18} .

Instructional Tasks

Instructional Task 1 (MTR.6.1)

The diameter of fishing lines varies. Fishing lines can have a diameter as small as 2×10^{-2} inch and as large as 6×10^{-2} inch.

Part A. Which value belongs to the thicker fishing line?

Part B. How many times larger is the thick line compared to the thin line?

Part C. If you want a fishing line whose thickness is in between the two values, what would be a possible thickness for the line you would like to use?

Instructional Task 2 (MTR.6.1)

The state of Florida is approximately 6.5×10^4 square miles. Lake Okeechobee is the largest freshwater lake in Florida and covers approximately 7.3×10^2 square miles.

Part A. Is it reasonable to estimate Florida to be 100 times larger than Lake Okeechobee? Why or why not?

Part B. How many times larger is Florida than Lake Okeechobee?

Instructional Task 3 (MTR.6.1)

In 2020, three of the countries with the highest population included China, the United States and Mexico. The population of the United States was 331.45 million. The population of China was 1.44 billion. The population of Mexico was 129.2 million.

Part A. What would be an appropriate estimate for the number of times larger China is compared to the United States? Compared to Mexico?

Part B. How many times larger is China than the United States?

Part C. How many times larger is China than Mexico?

Instructional Items*Instructional Item 1*

The distance in kilometers to Proxima Centauri, the closest star to Earth, is 39,900,000,000,000. Estimate the distance in kilometers to Proxima Centauri by writing it in the form of a single digit times an integer power of 10.

Instructional Item 2

The Bohr radius of a hydrogen atom is 0.000000000529. Express the Bohr radius of a hydrogen atom in scientific notation.

Instructional Item 3

The average weight of a blue whale is 4×10^5 pounds. The average weight of an elephant is 1×10^4 pounds. Approximately how many times heavier is a blue whale than an elephant in pounds?

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*